

# Introduction to Linux OS and Bash shell

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(E62, Dense and Strange Hadronic Matter)



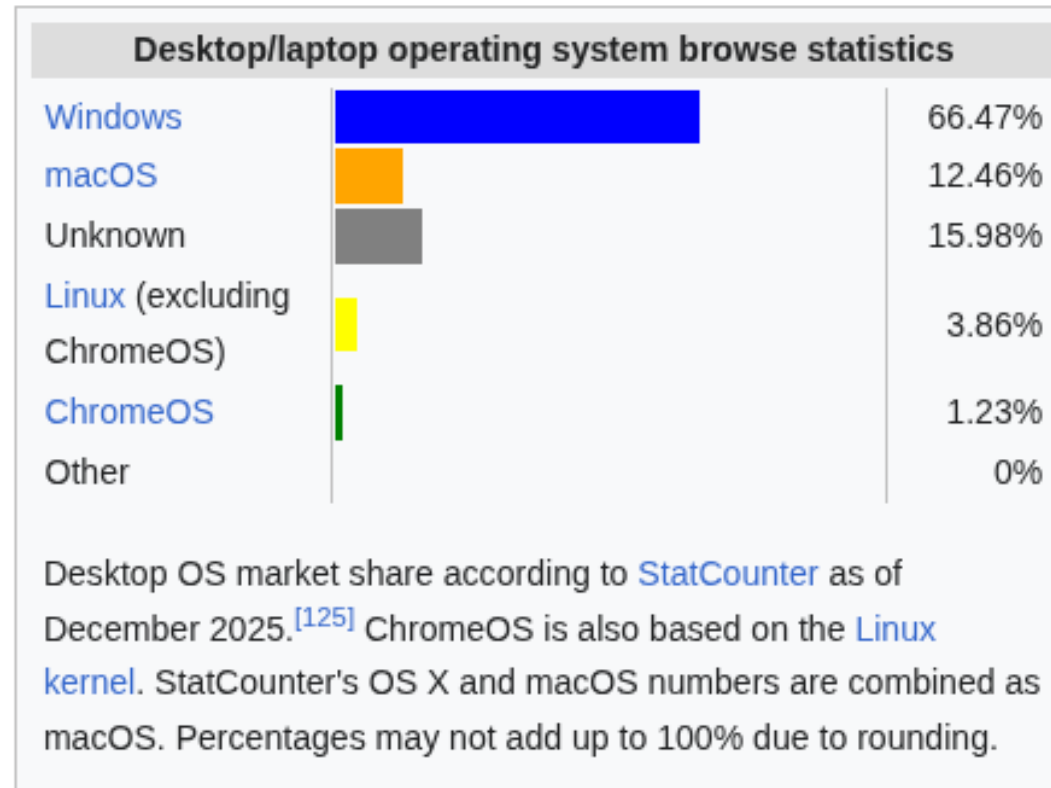
# Outline

- Introduction
  - Linux OS
  - Bash shell



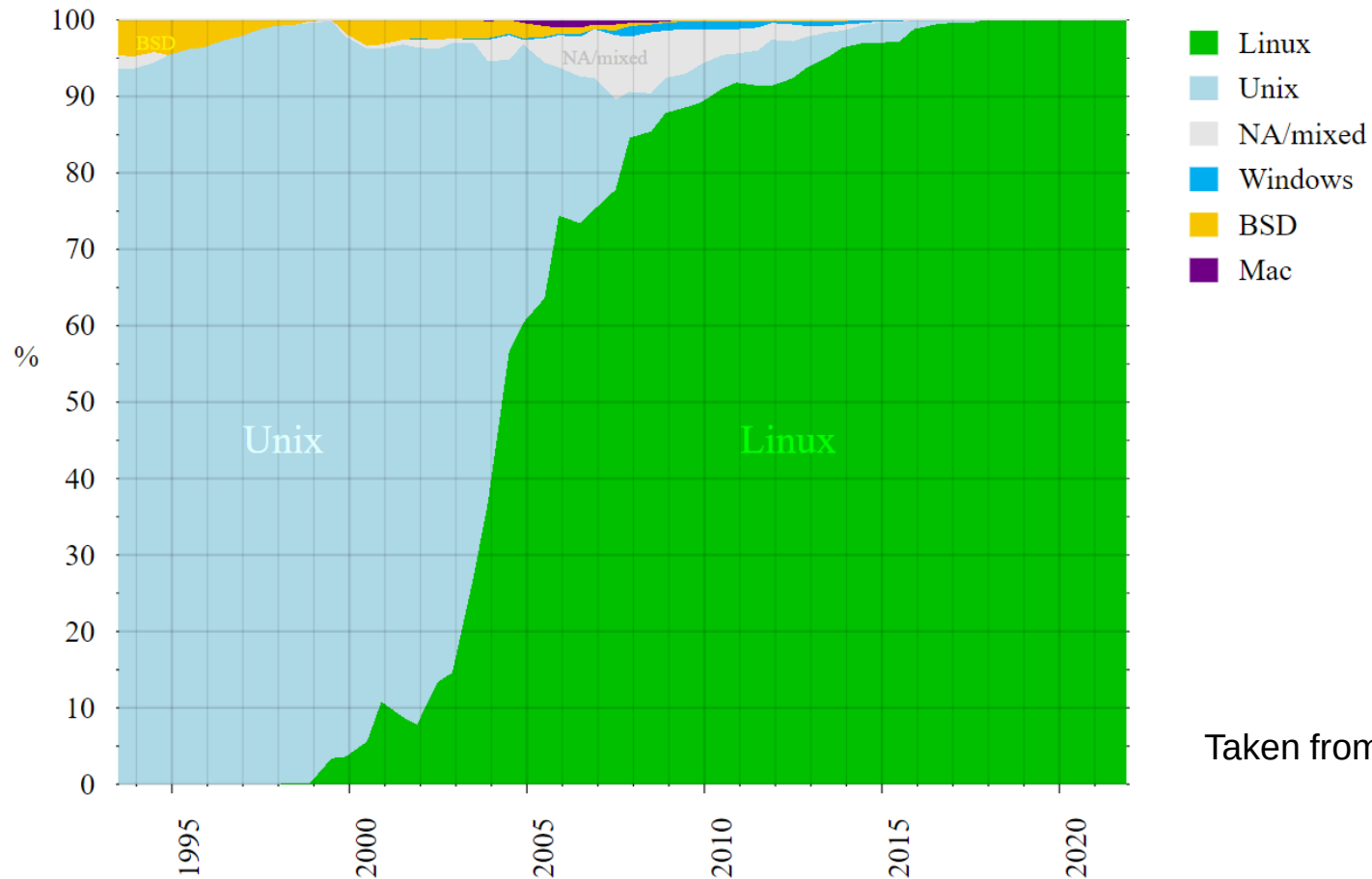
# Why Linux OS?

- Statistics for all desktop/laptop computers:



# Why Linux OS?

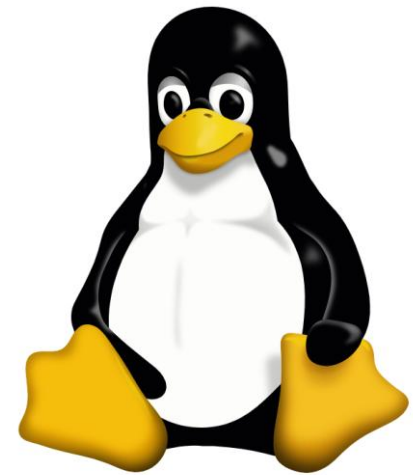
- Statistics for all supercomputers:



Taken from this [link](#)

# Why Linux OS?

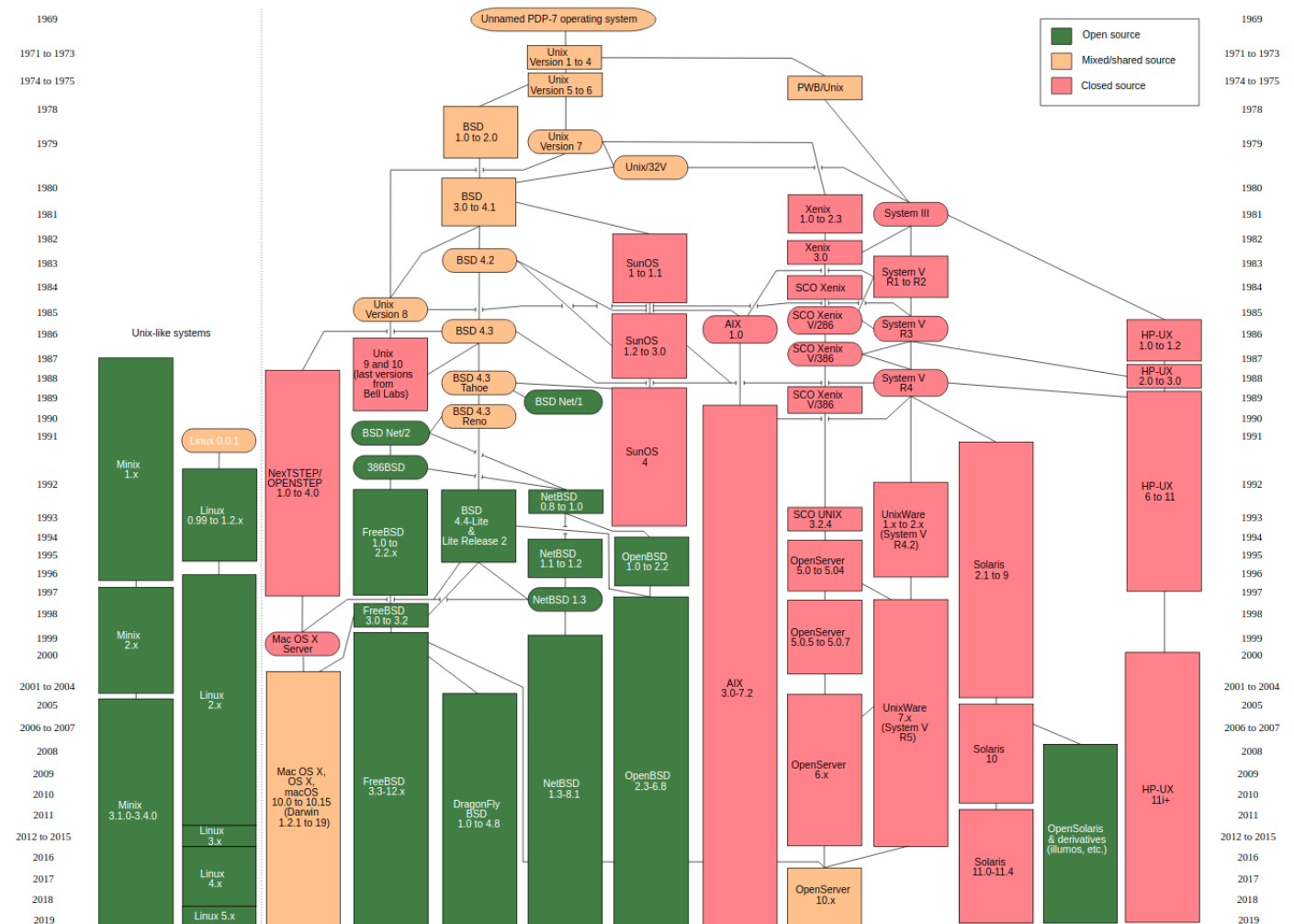
- Linux is by far the leading operating system in computers used in scientific research (CERN, NASA, etc.)
- **Linux kernel:** developed by Linus Torvalds in the early 90s
- **GNU ('GNU's not Unix'):** open source utilities to perform standard actions on the computer – no licensing
  - Linux kernel + GNU utilities = Linux OS
- Initially, the Linux OS was basically a free Unix



Penguin named **Tux** is the most commonly used logo for Linux

# Unix OS

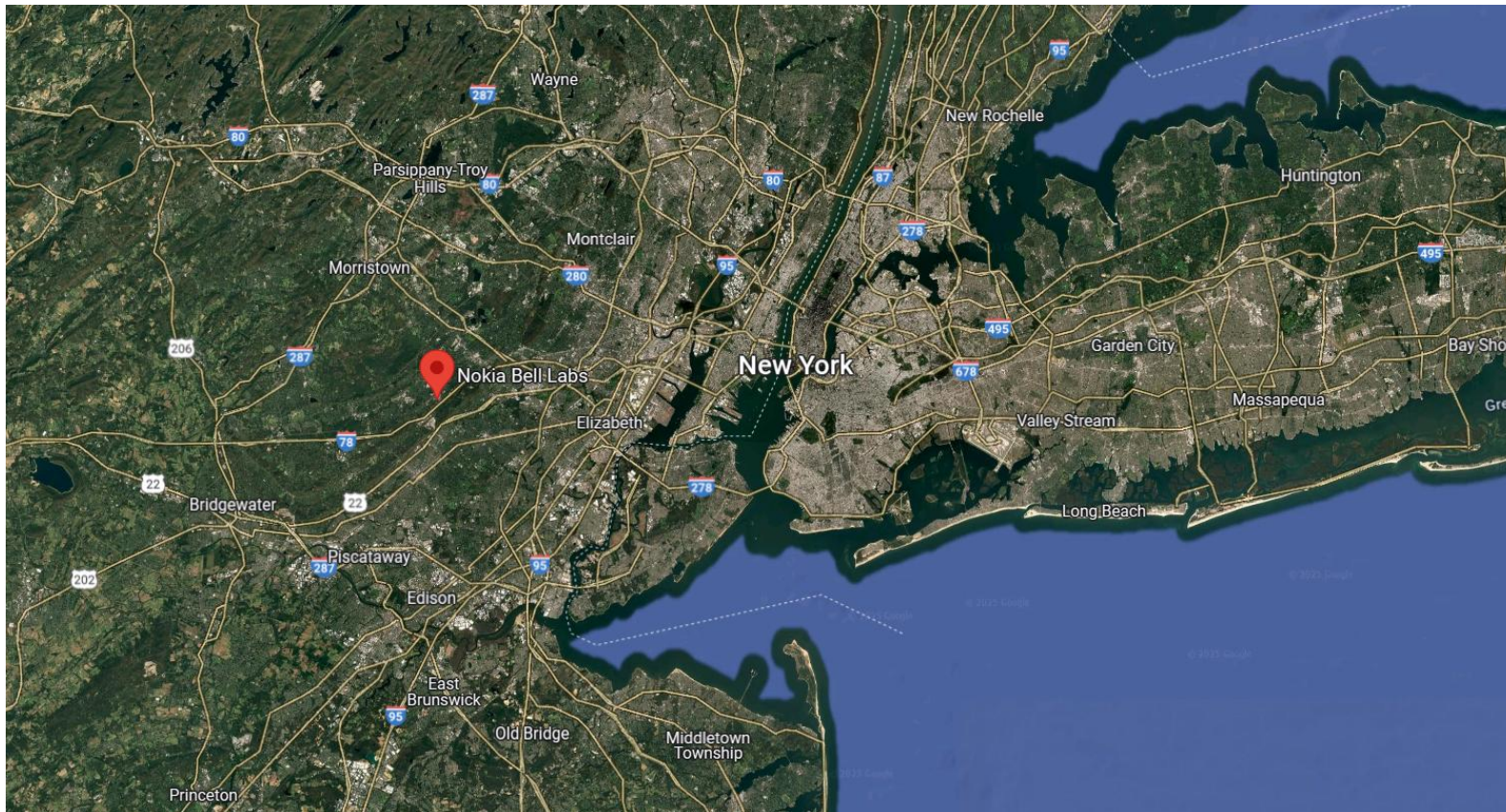
- Unix legacy:
  - The first OS written in a high-level programming language (C)
  - The first widely portable OS Unix v7 (1979)
- Unix family tree – Linux is one of its direct descendants
  - Mac OS is also part of this family!





# Unix OS

- Unix was developed at “Bell Labs” in Murray Hill, New Jersey, USA, in the 70’s



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# Linux OS

- Linux family tree: Common Linux kernel and a plethora of different Linux distributions built on top of it
  - Full-core: Debian, Red Hat Enterprise, openSUSE, etc.
  - Specific: Ubuntu, Fedora, CentOS, Linux Mint, etc.
- Specific distributions are derivatives of full-core distributions:
  - Ubuntu is derivative of Debian, Fedora of Red Hat, etc. (see complete tree at <https://distrowatch.com/images/other/distro-family-tree.png>)
- The material presented in this course will be demonstrated on Ubuntu, but it applies also to any other Linux distribution
- Regular updates on all distributions: <https://distrowatch.com>

The Ubuntu logo, featuring the word "ubuntu" in white lowercase letters on an orange rectangular background, with a small white circular icon containing a gear-like pattern to the right.

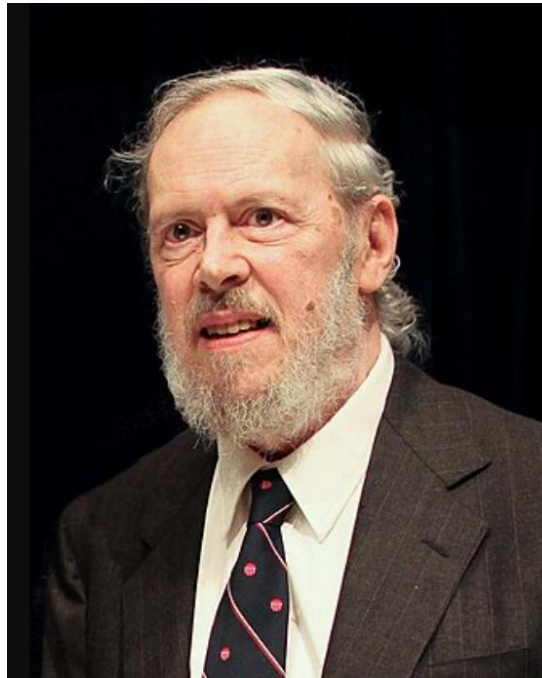
# Key people

- Ken Thompson



B programming language,  
UNIX operating system,  
Regular expressions,  
UTF-8 encoding,  
Go programming language,  
Chess tablebases, ...

- Denis Ritchie



C programming language,  
UNIX operating system,  
...

- Linus Torvalds



Linux operating system,  
Git (version control system),  
...

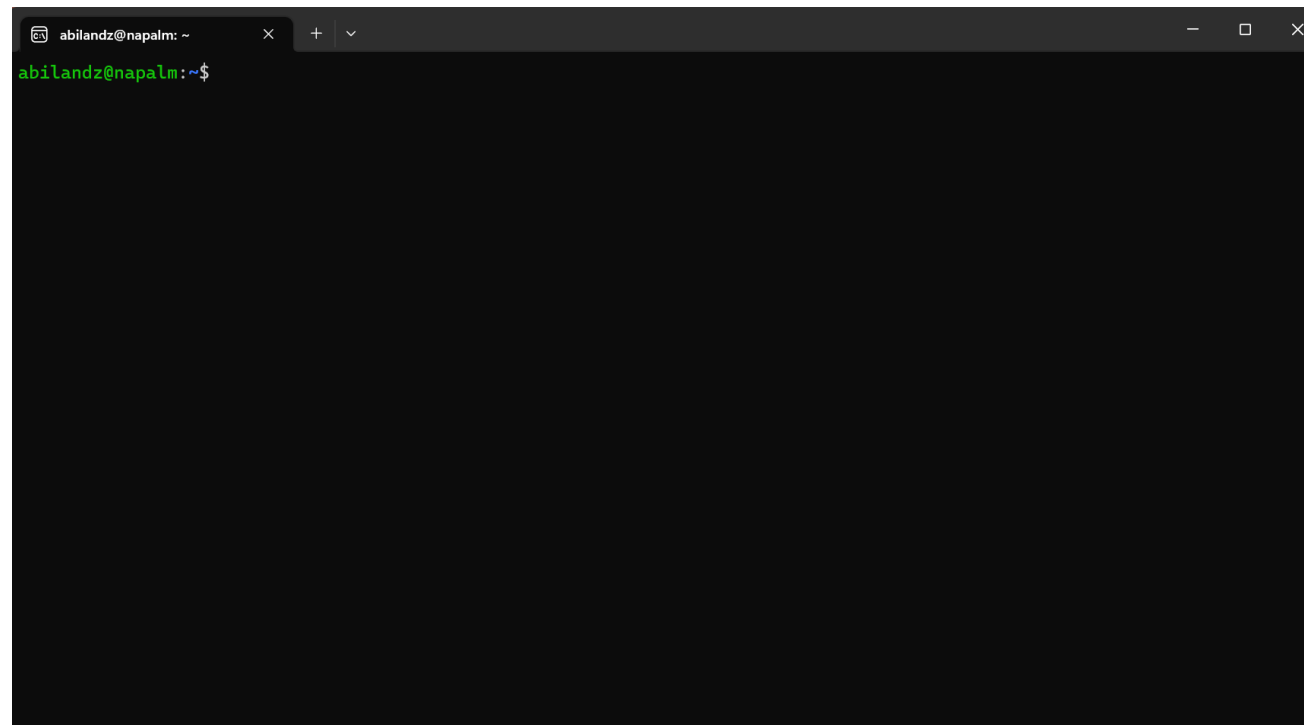
# Getting Ubuntu for Windows users



- If you have only Windows on your laptop, you could either:
  - on Windows 11, use [Windows Subsystem for Linux](#) :
    - Open Windows PowerShell and list available Linux distributions with the command:  
**wsl --list --online**
    - Pick up your favorite Linux distribution, and install it e.g. as follows:  
**wsl --install Ubuntu-24.04**
    - The rest is easy (pick up your account, etc.)
- or, install PuTTY ( <https://www.putty.org/> ) and then use it to connect and work remotely on some computer running Linux, on which you have an account

# Is this the right crash course for you?

- If, after you have opened a terminal in Linux ...



... you have asked yourself: 'What now?', then this is the right course for you!



# Is this the right crash course for you?

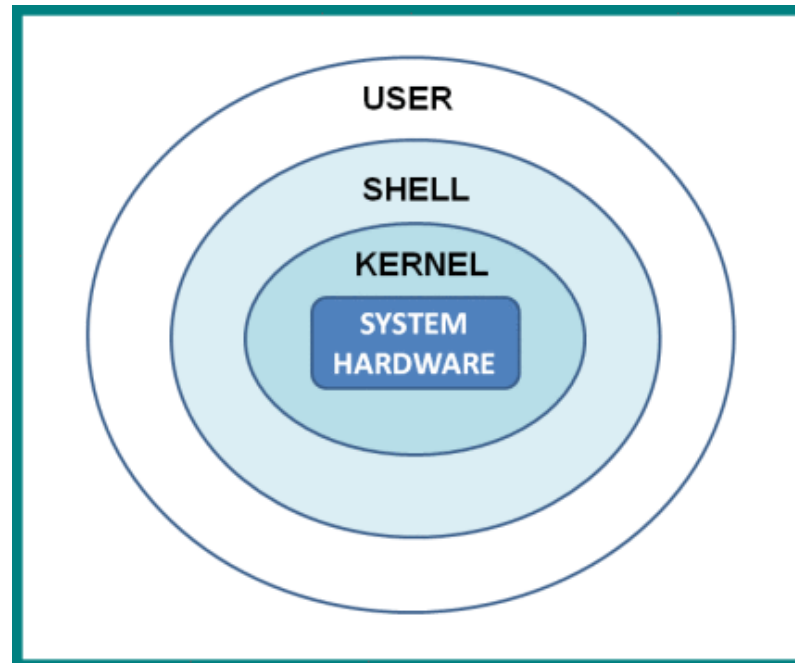
- What can we do in the terminal?
  - Not that much with the mouse...
- Next, you can start typing and pressing 'Enter', but most likely, whatever you have typed in the terminal will produce only error messages
  - Still, that is something, as it clearly means that there is some secret/magic language that is trying to respond to, or to interpret, your command input, as soon as you have typed something in the terminal and pressed 'Enter'. What is that secret built-in language available in the terminal?

# Linux shells

- Loosely speaking, a shell is any program that a user employs to type commands in the terminal (text window)
- Example shells:
  - sh
  - bash
  - zsh
  - ksh
  - csh
  - fish
  - PowerShell (developed by Microsoft)
- Since Bash is the default shell on most Linux distributions nowadays, we focus on it
  - If not set by default, just type **bash** in the terminal, and you are in the Bash wonderland

# Why shell?

- The shell translates the commands you type into a format that the computer can understand
  - It works both ways: The user is shielded from the Linux kernel, and the Linux kernel is shielded from the user



# A bit of Bash history

- Initial development by Brian Fox in 1989 as a part of GNU project...
  - And it's still alive!
- Bash is an acronym for 'Bourne-again shell'
  - The original shell 'sh' was written in 1977 by Stephen Bourne
- Written entirely in C
  - The Linux kernel is mostly written in C
- Executable: /bin/bash
- File extension: .sh
- Command processor / interpreted / scripting language



# The current status of Bash

- Bash is well-maintained and is still under regular development
  - Webpage: <https://www.gnu.org/software/bash/>
  - Source code: <http://git.savannah.gnu.org/cgiit/bash.git>
- Latest release: version 5.3 (July 3rd, 2025)
  - The current maintainer: Chet Ramey





# Interpreted vs. compiled languages

- Interpreted:
  - write code & execute line-by-line
  - less reliable (there is no compiler to catch the errors!)
  - the source code can be easily read and copied
  - examples: Bash, Python, Mathematica, JavaScript
- Compiled:
  - write code & compile & execute the compiled file ('binaries')
  - generally runs faster than interpreted code
  - examples: C, C++, Java, Go

**Thanks!**